Definite Integrals From the Marriage of Power and Fourier Series

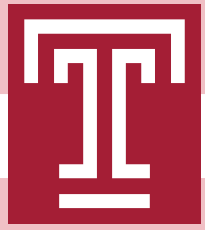
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EPaDel Section Meeting

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Part I

Introduction

Basic Idea

- Goal: Generate a table of definite integrals.
- Integrals of the form:

$$\int_{-\pi}^{\pi} f(x)\,dx, \quad \int_{-\pi}^{\pi} f(x)\cos(nx)\,dx, \quad \int_{-\pi}^{\pi} f(x)\sin(nx)\,dx.$$

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- How: Use Taylor Series and Fourier Series.
- Why: Another technique to obtain exact solutions to non-trivial integrals, instead of resorting to approximation methods.
- End Result: We generate a table of several definite integrals.

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How can we evaluate the integral below?

$$\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\cos(n\theta)\ d\theta.$$

We can use Taylor series and Fourier Series to help us.

General Process

1. Start with a given function $g(z)$.
2. Substitute $z = re^{i\theta}$, rule: $e^{i\theta} = \cos\theta + i\sin\theta$.

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5. Equate Real with Real and Imaginary with Imaginary.
6. Use Fourier Series Definition.

Taylor Series

Theorem

If f has a power series expansion at a, that is, if

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n \quad |x-a| < R,$$

then the coefficients are given by $c_n = \frac{f^{(n)}(a)}{n!}$. Hence,

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a)^1 + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

Fourier Series

Definition

If $f \in L[-\pi, \pi]$, then the Fourier series for f is given by

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(kx) + \sum_{k=1}^{\infty} b_k \sin(kx) \text{ for } -\pi \leq x \leq \pi.$$

Where, the *Fourier coefficients* are given by

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(kx)\, dx \text{ for } k = 0, 1, 2, 3, \ldots, \quad (1)$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(kx)\, dx \text{ for } k = 1, 2, 3, \ldots \quad (2)$$

A Crazy Integral

- We begin with a given function and we find a closed formula for a non-trivial definite integral.
- $$\int_{-\pi}^{\pi} \cos(r\cos\theta)\cosh(r\sin\theta)\cos(2n\theta)\ d\theta = \frac{(-1)^n \pi r^{2n}}{(2n)!}$$

Part II

Step By Step Example

Given $g(z)$, write $g(z) = u(r, \theta) + iv(r, \theta)$.

Let $g(z) = e^z$, then substitute $z = re^{i\theta}$.

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$$e^z = \underbrace{e^{r\cos\theta}\cos(r\sin\theta)}_{u(r,\theta)} + i\underbrace{e^{r\cos\theta}\sin(r\sin\theta)}_{v(r,\theta)}$$

We have decomposed g into a real and imaginary part.

Write the Taylor Series as Real + Imaginary

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$$= \sum_{n=0}^{\infty} \frac{r^n}{n!} \left(\cos(n\theta) + i\sin(n\theta)\right)$$

$$e^z = 1 + \sum_{n=1}^{\infty} \frac{r^n}{n!} \cos(n\theta) + i \sum_{n=1}^{\infty} \frac{r^n}{n!} \sin(n\theta)$$

This is a Fourier Series Expansion of e^z.

Match Up the Real and Imaginary Parts

$$g(z) = e^z \quad = \quad u(r,\theta) \quad + \quad iv(r,\theta)$$

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Then equate real and imaginary parts to get

$$e^{r\cos\theta}\cos(r\sin\theta) = 1 + \sum_{n=1}^{\infty}\frac{r^n}{n!}\cos(n\theta)$$

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$$\frac{r^n}{n!} = \frac{1}{\pi}\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\cos(n\theta)\ d\theta \text{ and}$$

$$\frac{a_0}{2} = 1 = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\ d\theta$$

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$$\begin{aligned} e^{r\cos\theta}\sin(r\sin\theta) &= \sum_{n=1}^{\infty}\frac{r^n}{n!}\sin(n\theta) \\ v(r,\theta) &= \sum_{n=1}^{\infty} b_n \sin(n\theta) \end{aligned}$$

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If we isolate the integrals from the previous slides we get for $n = 1, 2, 3, \ldots$ and $r \in \mathbb{R}$

- $$\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\ d\theta = 2\pi$$

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We can repeat the procedure for $g(z) = \cos z$ to get

- $$\int_{-\pi}^{\pi} \cos(r\cos\theta)\cosh(r\sin\theta)\ d\theta = 2\pi$$

- $$\int_{-\pi}^{\pi} \cos(r\cos\theta)\cosh(r\sin\theta)\cos(2n\theta)\ d\theta = \frac{(-1)^n \pi r^{2n}}{(2n)!}$$

- $$\int_{-\pi}^{\pi} \sin(r\cos\theta)\sinh(r\sin\theta)\sin(2n\theta)\ d\theta = \frac{(-1)^{n+1} \pi r^{2n}}{(2n)!}$$

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- Used procedure to obtain 36 integrals.
- Most numerically verified, but some maybe new integrals.
- One challenge: being able to decompose $g(z) = u(r, \theta) + iv(r, \theta)$ easily.
- Generates closed formulas for integrals over $[-\pi, \pi]$.
- Need to keep track of the radius of convergence.

References

[1] Y. Brychov.
Hand Book of Special Functions: Derivatives, Integrals, Series, and Other Formulas.
CRC Press, 2008.

[2] I.S. Gradshteyn and I.M. Ryshik.
Table of Integrals, Series and Products.
Academic Press,2nd Edition, 1980.

[3] G.P. Tolstov.
Fourier Series, (Translated from the Russian by Richard A. Silverman).
Dover Publications, 1962.